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Stratigraphic Relations of Members of the Needles Range Formation at Type Localities in Southwestern Utah

ABSTRACT

The Needles Range Formation is one of the most extensive of the middle Tertiary ash-flow sheets in the Great Basin of Nevada and western Utah. The original definition of the formation by Mackin (1960) named two members and mentioned three type localities. Our recent studies have shown that this published information now requires revision. To provide a firm basis for stratigraphic studies of the Needles Range Formation, this paper presents geologic maps, measured sections, and paleomagnetic data for three type localities in southwestern Utah. These include the northern Needles Range and Wah Wah Springs localities previously mentioned by Mackin, plus a new locality, northwest of Lund, Utah. The sequence of members at these localities, from oldest to youngest, is: Cottonwood Wash Tuff (new name), Wah Wah Springs Tuff, Lund Tuff (new name), and Wallaces Peak Tuff (new name). We recommend abandonment of the name "Minersville Tuff."

INTRODUCTION

The Needles Range Formation is one of the most extensive and lithologically distinctive of the numerous Tertiary ash-flow deposits in the Great Basin. Known exposures extend over an area of more than 33,600 sq km (13,000 sq mi) in southwestern Utah and eastern Nevada. Abundant phenocrysts of plagioclase, biotite, and amphibole distinguish the unit from older and younger, generally more rhyolitic tuffs. These aspects, together with plentiful K-Ar dates (Armstrong, 1970) which establish an age of 29.7 ± 0.9 m.y., make the Needles Range Formation a key stratigraphic unit in the succession of Tertiary ash-flow deposits in the region.

The formation was defined by Mackin (1960, 1963) as a sequence of usually pink to red-brown crystal-rich tuffs with the type locality found in the northeastern arm of the Needles Range near the Garrison-Milford Highway (Figs. 1 and 2). Mackin further stated (1960, p. 99): "The lower of the two members at the type locality is here designated the Wah Wah Springs Tuff, for its occurrence just south of Wah Wah Springs . . . at this place the unit is 700–800 feet thick . . . the upper member at the type locality is designated the Minersville Tuff for its occurrence in the

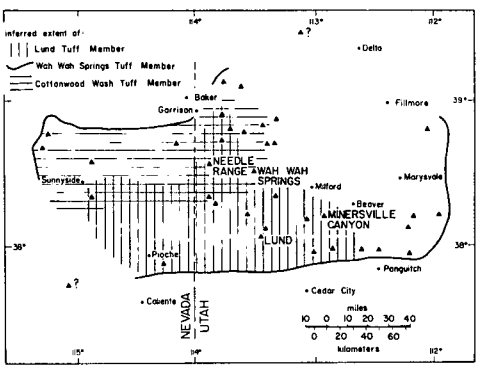


Figure 1. Index map for the Needles Range Formation. Investigated localities, all having exposed Wah Wah Springs Tuff, are shown by solid triangles, with the type localities labeled in bold letters. Western limits of the formation partly are after Grommé and others (1972).

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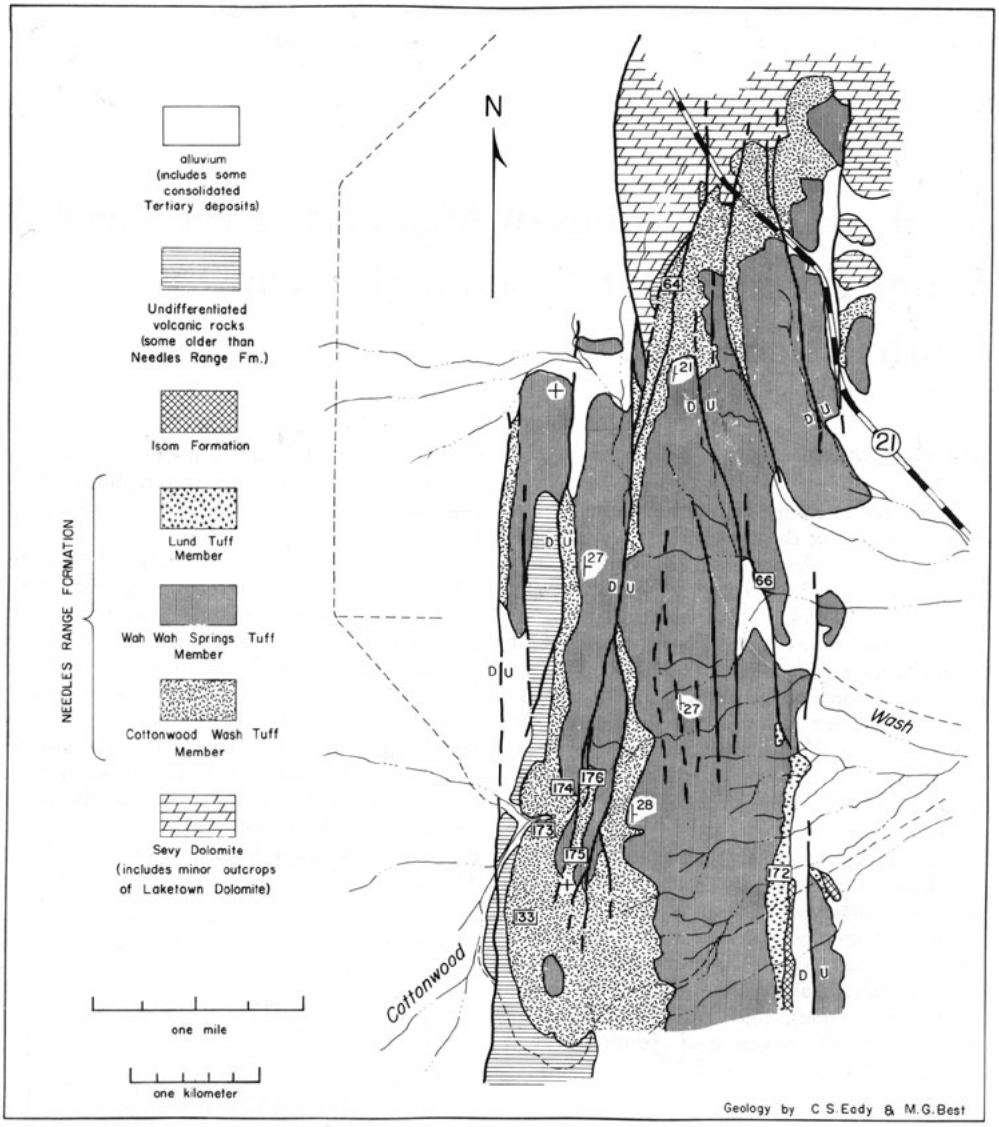


Figure 2. Geologic map of the northeastern arm of the Needles Range where it is crossed by the Garrison-Milford Highway. Numbers in boxes refer to paleomagnetic sample sites (Table 1). Section corners are shown as crosses. The type locality for the Cottonwood Wash Tuff Member is in Cottonwood Wash at sample site 173 on the east side of sec. 4, T. 26 S., R. 18 W.

Minersville Canyon of the Beaver River. . . ." Subsequent investigators have either accepted (Cook, 1965; Armstrong, 1970) or have experienced difficulty in applying (Conrad, 1969; Grommé and others, 1972) Mackin's stratigraphy.

Recent mapping at these type localities has disclosed serious discrepancies in the published relations. In the northeastern Needles Range, the Wah Wah Springs Tuff is the second, not the basal, unit within the formation. In the large prominent conical hill just south of Wah Wah Springs, the thick section of tuff referred to by Mackin consists of not one but three ash-flow tuffs of Needles Range character.

Exposures in Minersville Canyon consist of

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a complex sequence of badly faulted and altered volcanic rocks whose stratigraphic relations are obscure.

The aim of this note is to clarify the stratigraphic relations of the members of the Needles Range Formation and to name new units.

NORTHEASTERN NEEDLES RANGE[1]

Mapping by students in the Brigham Young University Summer Field Course on a scale of 1:20,000 (Fig. 2) has refined the structural and stratigraphic relations shown by Conrad (1969) at a scale of 1:62,500. Throughout most of the northeastern segment of the range, two ash-flow units of the Needles Range Formation are exposed, with a third found locally. We propose to designate the lowest depositional unit the Cottonwood Wash Tuff Member, named from its easily accessible exposures where Cottonwood Wash cuts across the western side of the northeastern arm of the range. This is the "gross biotite" unit of Conrad (1969). As implied by his name, a characteristic aspect is the presence of large books of biotite, commonly 6 mm across. Another useful distinguishing characteristic (Fig. 3) is the fact that amphibole is not conspicuous in hand specimen, although small (<1 mm) euhedra are seen in thin section. Also, in every hand specimen, at least one or two large embayed and (or) broken glassy quartz phenocrysts, not uncommonly 4 mm across, are readily found. Flattened lapilli and blocks of pumice are evident in the lower portion of the unit. A basal gray vitrophyre is locally developed. Throughout most of the unit the tuff is medium-red brown, moderately welded and compacted, and only poorly foliated. This generally massive fabric, combined with the abundance of large phenocrysts, especially plagioclase, creates a granite-like texture. The exposed top of the unit is pink, quite soft, and porous.

Overlying the Cottonwood Wash Tuff Member is the Wah Wah Springs Tuff Member. Wherever exposed in or near the Needles Range, the base of the Wah Wah Springs Member is a well-exposed layer of intensely welded black vitrophyre. Pumice lapilli are more abundant here than in the underlying Cottonwood Wash Tuff, and the average crystal size is less. Amphiboles are more abundant and larger in size so that they are invariably conspicuous in hand specimen. Quartz is generally not recognizable in hand specimen. Fairly large (3 to 4 mm) books of biotite may be seen in some exposures, but the obvious amphibole serves to distinguish it from the Cottonwood Wash Tuff Member.

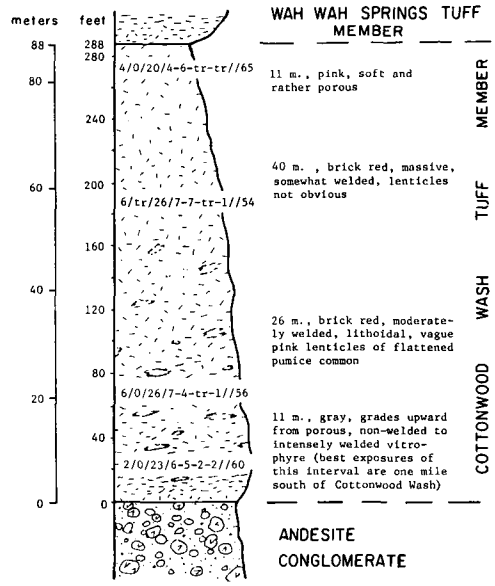


Figure 3. Section through the Cottonwood Wash Tuff Member at its type locality on the side of sec. 4, T. 26 S., R. 18 W. The modal percentages of constituents as determined in thin section are: quartz / sanidine / plagioclase / biotite / amphibole / pyroxene / Fe-Ti oxides // matrix.

Very locally, the Lund Tuff Member overlies the Wah Wah Springs Tuff. The Lund Tuff Member superficially differs only in the conspicuous presence of abundant, large, quartz phenocrysts.

In the eastern part of the map area (Fig. 2), the tuffs are extensively and intensively oxidized with development of abundant hematite. This alteration seems in many instances to be associated with intense shearing and brecciation in several fault zones.

WAH WAH SPRINGS

In contrast to the Needles Range where there are very numerous north-south–striking normal faults, the section around Wah Wah Springs (Fig. 4) is devoid of faults and forms a gentle east-dipping homocline. South of the springs, in the large prominent conical hill which rises about 300 m above the surrounding

[1] Designated by Mackin (1960) and Conrad (1969) as the Needles Range but shown as the Needle Range on the Richfield AMS topographic map.

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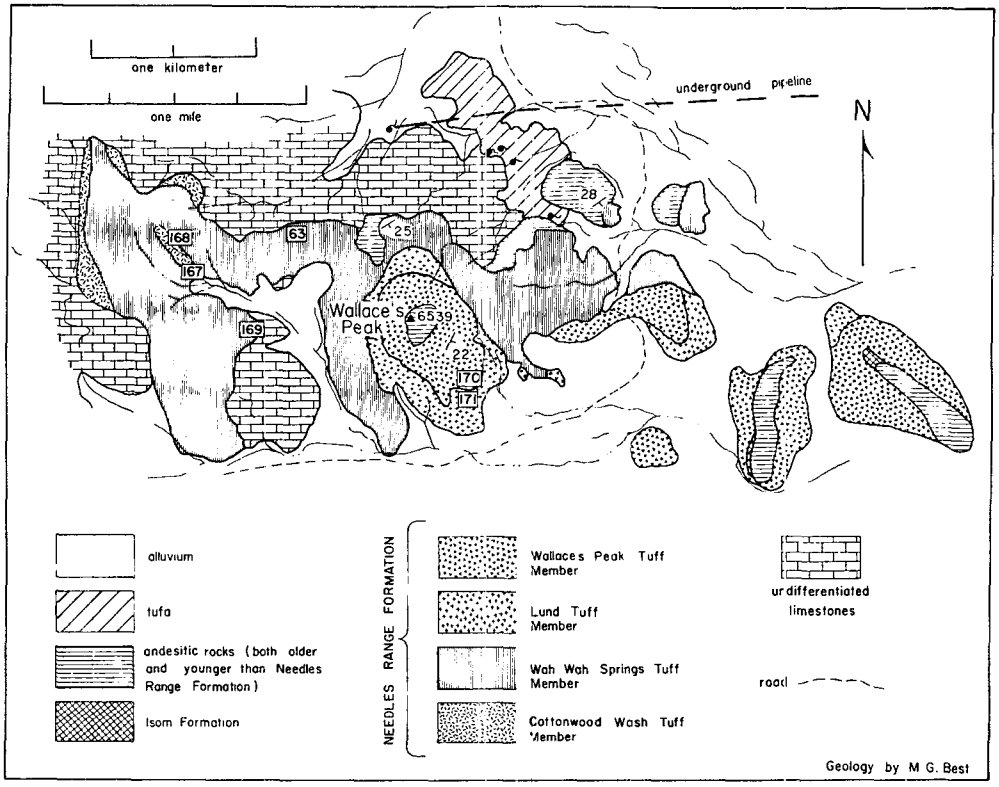


Figure 4. Geologic map of the Wah Wah Springs area. The Garrison-Milford Highway lies to the north just beyond the map area. Wallaces Peak is centered in the southeast quarter of sec. 11, T. 27 S., R. 15 W.

alluvial slope, there are three tuffs of Needles Range character, capped by a very thin wedge of Isom Formation and then by dark-colored lava flows. The older Cottonwood Wash Tuff Member is exposed in nearby side canyons.

The well-exposed tuff around the north base of Wallaces Peak, as the conical hill is locally known, is apparently the one intended by Mackin to be the Wah Wah Springs Tuff. Only in the western part of the map area, (Fig. 4), is the dense *black* vitrophyre (so typical of this unit in the ranges to the west into Nevada) well developed. Throughout most of the unit, light-colored lapilli of pumice are apparent in the darker red-brown matrix. Biotite, amphibole, and plagioclase are all conspicuous to the naked eye.

Overlying the Wah Wah Springs Tuff Member, around all but the east side of Wallaces Peak, is the Lund Tuff Member (Fig. 5). Throughout its entire thickness, which is generally less than 15 m, it is porous and poorly welded. Color ranges from light gray to buff. Apatite and sphene are evident in thin sections.

The steeper upper slopes of Wallaces Peak and parts of the hills to the east contain another unit more than 60 m thick, herein named "Wallaces Peak Tuff Member." That this is indeed a separate depositional unit may be seen in massive ledges southeast of the peak where an interval less than 1 m thick of somewhat sorted and stratified tuff separates the unit from the underlying Lund Tuff. As in the latter, welding and compaction are slight, and color is medium gray. Unlike the Lund, however, prominent white pumice lapilli comprise 10 to 20 percent of the tuff and quartz phenocrysts (generally <3 mm) are only sparsely visible in hand specimen. Amphibole and biotite phenocrysts are about the same size as the quartz. In contrast to the Wah Wah Springs Tuff, the Wallaces Peak Tuff shows conspicuous zircon, apatite, and sphene in thin section.

LUND

The prominent volcanic section exposed (Fig. 6) on the ridge northwest of Lund, Utah,

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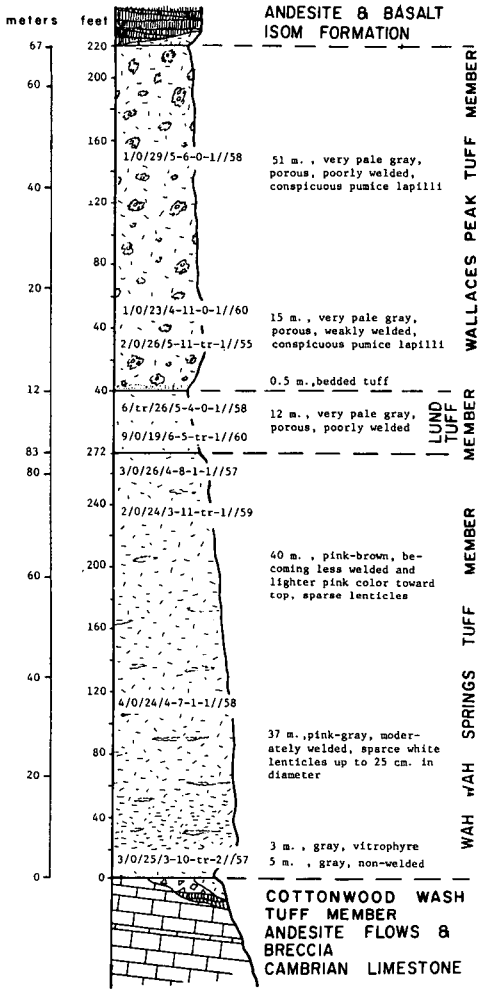


Figure 5. Section through the north side of Wallaces Peak south of Wah Wah Springs in sec. 11, T. 27 S., R. 15 W., the type locality for the Wah Wah Springs Tuff and Wallaces Peak Tuff Members.

includes rocks ranging from volcanic conglomerate predating the Needles Range Formation to the Isom Formation. Upward from the conglomerate, the sequence is: (1) locally exposed quartz-rich tuffs, the lowermost known informally as the tuff of Frisco; (2) several crystal-poor welded tuffs, the majority of which are informally called the tuffs of Escalante Valley and which form the lowest of three ledges that compose the main ridge in the map area; (3) Wah Wah Springs Tuff Member; (4) a thin complex of flows, breccia, sandstone, and crystal-poor welded tuff; (5) Lund Tuff Member; and (6) Isom Formation, consisting of several thin (<30 m) ash-flow tuffs and at least one lava flow. The volcanic rocks and conglomerate rest with apparent angular unconformity on what is presumed to be the Navajo Sandstone. The tuff of Frisco and tuffs of Escalante Valley below the Wah Wah Springs Tuff Member are not correlative petrographically or magnetically with the Cottonwood Wash Tuff even though they occupy a similar stratigraphic position. The thin complex of unnamed flows, breccia, sandstone, and crystal-poor welded tuff lying between the Wah Wah Springs Tuff and Lund Tuff Members is apparently localized at the Lund locality and does not appear at either Wallaces Peak or in the northern Needles Range. Until further mapping and stratigraphic studies are made in southwestern Utah, we feel it is not warranted to assign formal names to these units above and below the Wah Wah Springs Tuff Member. Also, we have not included any of these unnamed or informally designated tuffs in the Needles Range Formation because they do not fit the original definition of the formation as crystal-rich tuffs with abundant phenocrysts of plagioclase, amphibole, and biotite (Mackin, 1960).

The Wah Wah Springs Tuff changes composition vertically (Dinkel, 1969). In the lower, highly welded 120-m part of the unit which forms the middle row of ledges on the east flank of the ridge northeast of Lund, amphibole phenocrysts are large and abundant. Plagioclase also reaches maximum abundance and largest size near the middle of the unit. However, amphibole is so scarce in the upper 60 m that inclusion of this moderately welded portion with the basal portion could be questioned. Two welding discontinuities, traceable on the bold outcrops, may indicate minor complications in the otherwise simple cooling history.

The Lund Tuff Member, named for the nearby railroad town, may be divided into three gradational compositional zones that also correspond to welding and textural differences (Fig. 7). Apparently, many depositional units are involved, close in time, so that the entire member cooled as a single unit. Kreider (1970) has documented vertical variations in modal composition on a regional basis. The Lund Tuff Member is distinguished by large abundant quartz crystals, and by the presence of sphene in nearly every thin section. The average crystal percentages, obtained by counting more than 100,000 points in 12 samples from the type locality are: quartz, 5.45; alkali

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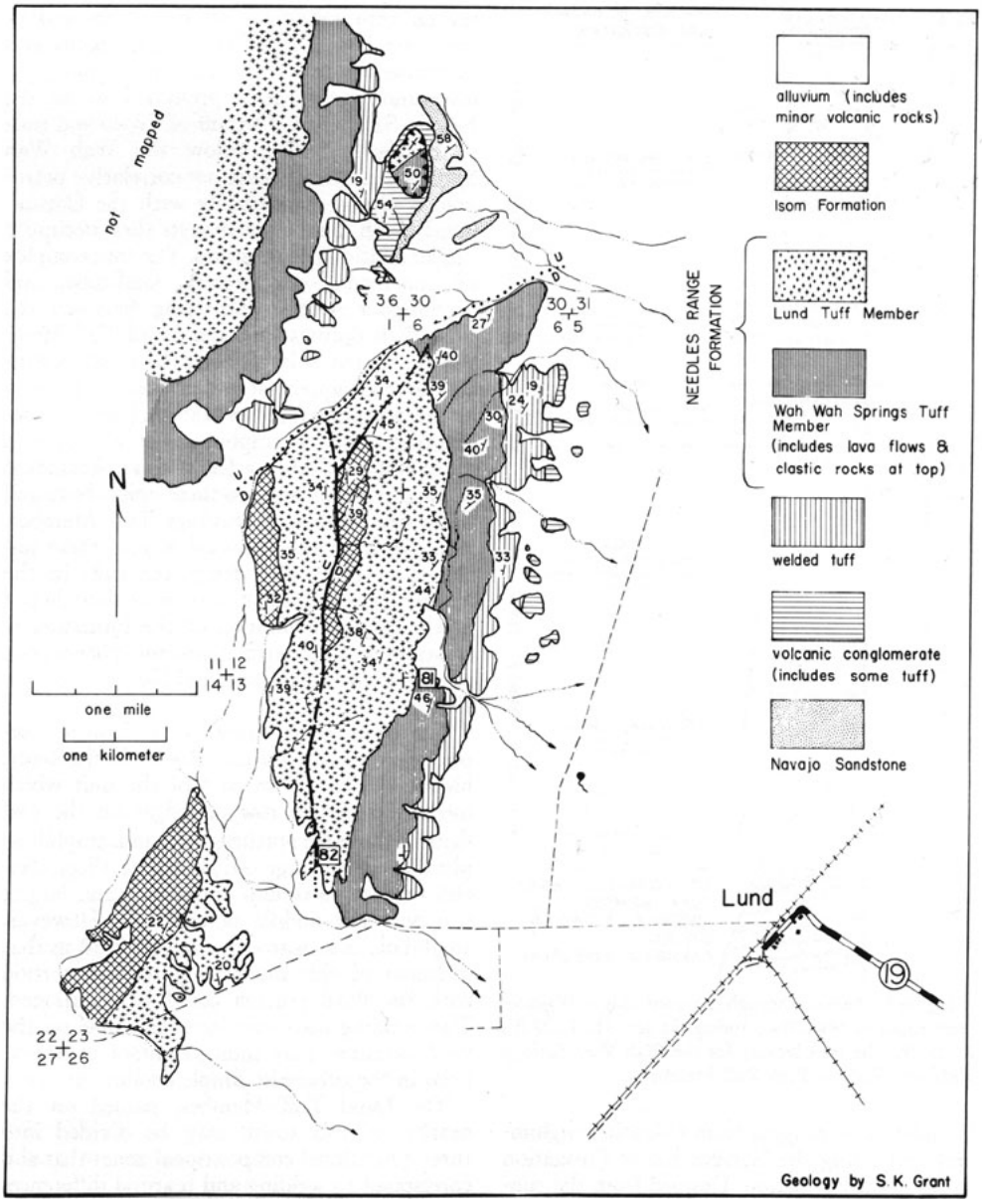


Figure 6. Geologic map of the area northwest of Lund, Utah, lying mostly within T. 32 S., R. 14 W.; and T. 32 S., R. 15 W. (note labeled section corners). An unnamed unit of lava flows and clastic rocks lying between the Wah Wah Springs Tuff and Lund Tuff Members has been included with the former member for cartographic convenience.

feldspar, 0.15; plagioclase, 22.76; biotite, 3.02; amphibole, 4.46; clinopyroxene, 0.12; opaque oxides, 0.87; sphene, 0.027; apatite, 0.017; and zircon, 0.0045. The largest quartz grain in each thin section averages 2.9 mm, plagioclase 2.3 mm, biotite 1.7 mm, and amphibole 1.2 mm.

PALEOMAGNETISM

An extensive paleomagnetic study of the middle Tertiary ash-flow sheets of the eastern Great Basin has been reported by Grommé and others (1972), who give references to

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earlier work. Our experience agrees with their conclusions concerning the magnetic characteristics of ash-flows. Partial alternating field demagnetization almost always reduces the 95 percent confidence radius for a given unit at a given site to less than the uncertainty in the correction for tectonic deformation. This uncertainty is typically 5° to 10°. All data are consistent with the hypothesis that the original direction of magnetization in an ash-flow sheet is uniform both vertically and laterally.

Our data on members of the Needles Range Formation at the type localities are presented in Table 1 and Figure 8.

The paleomagnetic data support the correlations made between the three type localities on the basis of petrography and stratigraphy. The four members may be distinguished from each other on the basis of magnetization direction. The Wah Wah Springs Tuff is the only member with wholly reverse polarity. The Wallaces Peak Tuff, although sampled only at its type locality, appears to be characterized by a low inclination and eastward declination. The Lund Tuff and Cottonwood Wash Tuff Members are close enough in magnetization direction that with the uncertainty in structural correction they would be difficult to identify on the basis of paleomagnetism alone. It does seem that the Lund Tuff has a steeper inclination and a more westerly declination than the Cottonwood Wash. Fortunately, these two members can be distinguished in the field on the basis of stratigraphic relations and mineralogy.

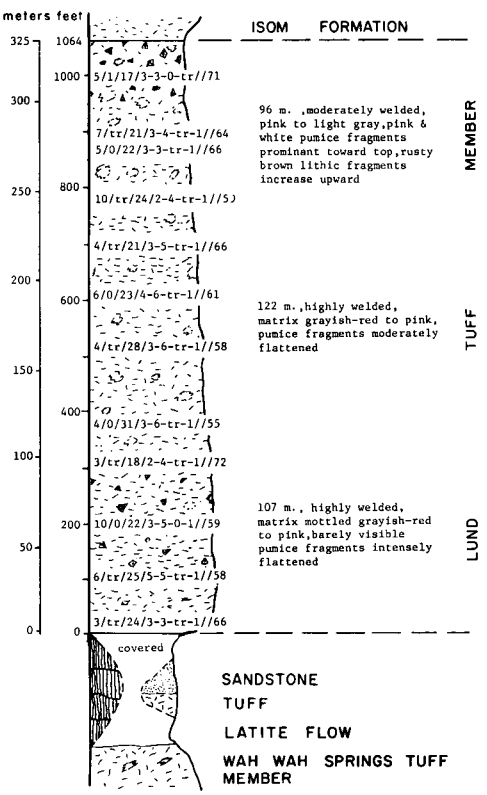


Figure 7. Section through the Lund Tuff Member at its type locality beginning in sec. 6, T. 32 S., R. 14 W., and extending into sec. 1, T. 32 S., R. 15 W.

At the Wah Wah Springs locality, only the upper part of the Cottonwood Wash Member is exposed (site 167). This is reversely magnetized like the overlying Wah Wah Springs Member and unlike the Cottonwood Wash Member in the northern Needles Range. To clarify the situation, additional samples of the Cottonwood Wash Tuff were collected at the type section, just below the base of the Wah Wah Springs Member (sites 175 and 176). These too were reversely magnetized. Thus we conclude that the top of the Cottonwood Wash Tuff is reversely magnetized over a wide area, presumably due to reheating by the overlying reversely magnetized Wah Wah Springs Tuff.

Paleomagnetic data on the Needles Range Formation have been previously reported by Grommé and others (1972). For the northern Needles Range, the directions indicated by Grommé for the two tuffs are completely consistent with our data. At the Wah Wah Springs locality, Grommé's drill holes were discovered in the Lund Tuff Member at the western base of the hill the farthest east of Wallaces Peak (Fig. 4). His reported direction is consistent (within the limits of uncertainty in the structural correction) with our data for the Lund Tuff Member, but not with the Wah Wah Springs Tuff Member.

MINERSVILLE PROBLEM

The detailed information given above has documented ambiguities and contradictions in the original definition of the Needles Range Formation by Mackin (1960). The ash-flow deposit he defined as the Wah Wah Springs Tuff actually consists of three petrographically and paleomagnetically distinct ash-flow tuffs. The unit apparently intended by Mackin as the Wah Wah Springs Tuff is actually the upper of the two units widespread in the northern Needles Range, not the lower as stated. This fact was recognized by Conrad

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TABLE 1. PALEOMAGNETIC DATA

Site No.	N	K	95	I	D	H
			Wallace's Peak Tuff			
170	5	190	7.1	32.	20.	300
			Lund Tuff			
82	8	520	2.4	64.	-36.	300
171	8	140	4.7	57.	-11.	300
172	9	270	3.2	66.	-29.	0
			Wah Wah Springs Tuff			
63	12	34	7.5	-32.	-170.	400
66	5	130	6.7	-47.	176.	300
81	8	210	3.8	-39.	172.	0
168	6	260	4.2	-31.	-169.	0
169	6	54	9.2	-36.	174.	300
174	6	44	10.2	-38.	-166.	300
			Cottonwood Wash Tuff, Top			
167	7	57	8.1	-27.	-174.	300
175	2	-	-	-8.	125.	300
176	2	-	-	-24.	173.	300
			Cottonwood Wash Tuff			
64	7	220	4.1	53.	-11.	300
133	7	27	11.8	29.	-11.	300
173	7	250	3.8	43.	-18.	0

Note: Site No., as shown on Figs. 2, 4, and 6; N, number of cores; k, Fisher precision parameter; 95, semi-angle of cone of 95% confidence of mean, degrees; I, D, inclination (positive downward) and declination (positive eastward) of mean NRM direction, degrees; H, peak alternating field in which most specimens were demagnetized, oersteds. At all sites at least two specimens were partially demagnetized in 300 oersted peak field.

(1969, p. 20) in a study conducted under Mackin. The application of the Wah Wah Springs name to the lower tuff in the northern Needles Range was clearly an error, because petrography and paleomagnetism prove they are different. Grommé and others (1972) attempted to use the Minersville name but immediately encountered difficulties.

The Minersville Tuff is defined by Mackin (1960, p. 99) as overlying the Wah Wah Springs Tuff in Minersville Canyon. This locality has been mapped by Erickson and Dasch (1968), and more recently by Rowley (1969, written commun.; see also Anderson, 1971). In Minersville Canyon itself, the rocks are altered and faulted; stratigraphic relations are sufficiently obscure that many experienced workers have been unable to identify the two Needles Range members on the basis of the information given by Mackin. His published data and oral communication to S. K. Grant (1966) indicate that the Minersville name was intended for the amphibole-rich, quartz-poor tuff exposed in Minersville Canyon and in the northern Needles Range. We find this tuff in both localities to be equivalent to the Wah Wah Springs Tuff at its type locality. Since Mackin has used two different names for the same tuff, we recommend abandonment of the Minersville name. Moreover, we prefer the Wah Wah Springs name because its type locality is clearly superior and the Minersville Canyon locality most inadequate.

Unpublished petrographic and paleomagnetic data obtained by us indicate that the two members of the Needles Range Formation mapped by Rowley throughout the area from Minersville Canyon southward 30 km are the Lund and Wah Wah Tuffs as defined in this paper. Although some might interpret that the unit we herein designate as the Lund is the unit which Mackin intended to call Minersville, we recommend complete abandonment of the Minersville name for a member of the Needles Range Formation because of the confusion in the use of this name in western Utah and eastern Nevada, and because of the unsuitability of Minersville Canyon as a type locality.

SUMMARY AND CONCLUSIONS

The Needles Range Formation consists of four separate, readily recognizable and map-

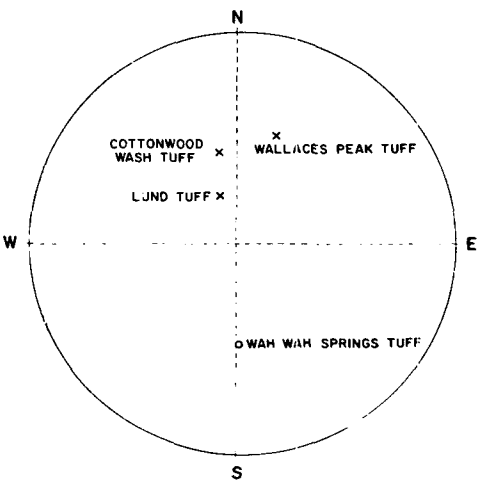


Figure 8. Mean NRM directions for the four named members of the Needles Range Formation. Equal-area projection, with cross for lower hemisphere and circle for upper hemisphere. Mean directions are computed from the data of Table 1, without including the remagnetized top of the Cottonwood Wash Tuff.

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pable members at the type localities. From bottom to top, they are: (1) Cottonwood Wash Tuff (new name) contains large biotites, sparse but conspicuously large quartz phenocrysts and abundant plagioclase; amphibole is generally not recognizable in hand specimen. (2) Wah Wah Springs Tuff contains conspicuous biotite, amphibole, and plagioclase. (3) Lund Tuff (new name) is characterized by abundant large quartz phenocrysts in addition to biotite, amphibole, and plagioclase. (4) Wallaces Peak Tuff (new name), occurring only at Wah Wah Springs, is characterized by more abundant pumice lapilli and fewer, smaller quartz phenocrysts than the Lund Tuff, which it otherwise resembles. Each of these four members has a distinct magnetic polarity, and the Wah Wah Springs Tuff is further distinguished by its reversed polarity.

The "700–800 foot thick" section of tuff in Wallaces Peak south of Wah Wah Springs consists of three Needles Range members—Wah Wah Springs Tuff overlain by Lund Tuff and Wallaces Peak Tuff.

Figure 1 illustrates our current understanding of the lateral extent of these members. The Wah Wah Springs Tuff is by far the most widespread and apparently is coextensive with all other members of the Needles Range Formation.

We can find no justification or need for retaining the name "Minersville Tuff" for a member of the Needles Range Formation and recommend it be abandoned.

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